



Hybrid Quantum-Classical Inverse Design of Metasurfaces for Tailored High Q-factor Response

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Abstract

The inverse design of metasurfaces is challenging due to complex interdependencies between structural parameters and electromagnetic responses, making traditional optimization methods computationally expensive and often suboptimal. This study employs a hybrid quantum-classical machine learning approach, known as the Latent Style-based Quantum GAN (LaSt-QGAN), to perform the inverse design of metasurfaces. This method integrates a Variational Autoencoder (VAE) with a Quantum Generative Adversarial Network (QGAN) to enhance the optimization of metasurface designs aimed at achieving narrow-band absorption. The proposed method results in a 6-fold reduction in training time and a 40-fold decrease in data requirements compared to traditional GAN-based approaches in both simulator and quantum hardware. The produced metasurface designs demonstrated a higher fidelity in relation to the target absorption spectra compared to the classical GAN method considered. Importantly, the proposed model generates metasurface designs with higher Q-factor ($Q=10^4$) than those trained on ($Q=10^3$). Additionally, the integration of a material lookup table facilitates manufacturability by allowing the substitution of predicted material properties with viable alternatives, while preserving spectral response.

Keywords Generative adversarial networks · Inverse design · Metasurface · Narrow-band absorption · Quantum computing · Quantum machine learning

1 Introduction

Metasurfaces are structures arranged in two-dimensional periodic arrays [1]. These structures exhibit unique optical properties, including narrow-band absorption, broadband absorption, beam steering, and unidirectionality, which enabled their widespread use in sensing, imaging, and telecommunications [2, 3]. Nonetheless, the design of metasurfaces to fulfill specific functional requirements remains a formidable challenge due to the non-uniqueness of solutions and the intricate interplay between their structural properties and electromagnetic responses [4]. Traditional methods for metasurface design often rely on extensive numerical

optimization or iterative trial-and-error techniques [5]. Although these approaches yield usable designs, they frequently demand substantial computing resources and may not reliably produce the most effective solutions [6].

To address these issues, inverse design has emerged as an efficient alternative, utilizing advanced algorithms to systematically determine the structural features necessary to achieve the desired optical properties [7]. In recent years, incorporation of artificial intelligence and machine learning into inverse design techniques has greatly enhanced the optimization procedures used to design metasurfaces [8–13].

Implementation of generative algorithms, particularly Generative Adversarial Networks (GANs), presents a significant advantage in this field, with extensive applications in image processing, video analysis, and molecular design [14]. The models presented in this study were formulated by training the encoded images, incorporating both the material and structural parameters. This framework facilitates the creation of metasurface designs that align with the designated target absorption spectra [15]. Despite

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these advancements, the intricate high-dimensional parameter spaces and complex design landscapes associated with metasurfaces continue to present major obstacles for conventional computational techniques. For instance, the occurrence of mode collapse, frequently observed in Generative Adversarial Networks (GANs), remains a critical issue [16, 17]. The computational requirements inherent in GANs are quickly approaching the limits defined by Moore's law. For instance, the process of generating images with dimensions 512×512 pixels through the application of GANs requires the handling of 158 million parameters. The procedure involves a training period of two days, utilizing a dataset that consists of 14 million instances, and is carried out on a setup of 512 tensor processing units (TPUs) [18].

Other notable methods using classical techniques like topology optimization, evolutionary algorithms, and deep-learning or neural-network-based inverse models face several fundamental limitations [19, 20]. Adjoint optimization, evolutionary search, and data-driven inverse models often require very large and dense datasets or many forward simulations to adequately cover the high-dimensional design space of both the geometry and material parameters. This makes the design expensive and computationally intensive, especially for metasurfaces requiring fine spectral/structural control [6, 8, 21]. Many machine-learning based inverse-design models generalize reliably mainly in the interpolation regime of the training data and can fail in extrapolation or out-of-distribution settings, which limit the discovery beyond the learned design space [22, 23]. While classical inverse design can optimize geometry (e.g., shape, periodicity, and meta-atom layout), the multi-objective optimization of both geometry and material properties such as dielectric constant, loss, and dispersion are challenging and is a key challenge when designing metasurfaces for realistic fabrication and functional performance [24–26]. Additionally, even among neural network-based methods, ensuring the physical validity and manufacturability of the generated designs is nontrivial. Some methods may produce non-physical parameter values (negative permittivity, impossible geometries) or designs that are difficult to fabricate [26, 27]. More importantly, the enormous complexity and nonlinearity of mapping from design parameters (geometry + material) to electromagnetic response (e.g., narrow-band absorption with high-Q resonances) leads to a design landscape that is high-dimensional, non-convex, and often consisting many local minima. Classical optimization and even classical GAN-based inverse design often struggle with such complexities, especially under data scarcity. [20, 26, 28, 29].

The landscape of inverse design has evolved rapidly recently, moving beyond standard discriminative networks toward more sophisticated generative paradigms. Recent articles and comprehensive reviews highlight a significant

shift towards physics-informed machine learning, where Maxwell's equations are directly embedded into the training loss to improve generalization in data-scarce regimes [30–32]. Furthermore, emerging architectures such as Denoising Diffusion Probabilistic Models (DDPMs) and Transformer-based photonic solvers have begun to outperform traditional GANs in generating high-diversity topological libraries [33, 34]. However, despite these classical advancements, the computational bottleneck of training on high-dimensional electromagnetic data remains a critical challenge, motivating the exploration of hybrid quantum-classical frameworks that leverage the improved expressibility and sampling efficiency of quantum circuits [35].

In this study, we address these limitations by implementing Quantum Generative Adversarial Networks (QGANs), which offers a novel approach to the inverse design of metasurfaces. These methods exhibit considerable potential in improving the design process through their ability to efficiently span and optimize parameter spaces [36–38]. Contemporary theoretical studies indicate that quantum generative algorithms may provide an exponential edge over classical algorithms, consequently generating increased interest in the theoretical and experimental aspects of quantum GANs [39–43].

While the application of quantum computing to electromagnetics is nascent, recent efforts have demonstrated the utility of quantum algorithms for specific photonic tasks. Variational Quantum Algorithms (VQAs), such as the Quantum Approximate Optimization Algorithm (QAOA), have been successfully applied to the parameter optimization of photonic circuits and networks [44, 45]. However, these approaches are restricted to parameter optimization, where the topology of the device is fixed, and the algorithm seeks optimal values for a discrete set of variables. They do not address the challenge of generative inverse design, where the goal is to discover entirely new, freeform geometries from a desired spectral response.

Concurrently, Quantum Generative Adversarial Networks (QGANs) have shown promise in generating synthetic data for standard benchmarking datasets like MNIST and financial modeling [39, 46, 47]. Yet, the direct application of pixel-based QGANs to nanophotonics is hindered by the high dimensionality of metasurface unit cells (64×64 pixels or higher), which far exceeds the qubit capacity of current Noisy Intermediate-Scale Quantum (NISQ) devices.

In this context, the novelty lies in bridging this gap. Unlike pure optimization algorithms, our framework is generative, capable of synthesizing novel topologies. Unlike standard QGANs, we introduce a hybrid architecture that employs a classical Variational Autoencoder (VAE) to compress the high-dimensional design space into a latent manifold. The quantum generator then operates as a latent-space

sampler, leveraging the high expressibility and entangling capabilities of quantum circuits to explore this compressed manifold more efficiently than classical counterparts. To the best of our knowledge, this represents the first application of a latent-style hybrid quantum generative model specifically tailored for the inverse design of high-Q metasurfaces.

The primary objective of GANs is to learn and approximate the underlying probability distribution of the training data [14]. This approach involves simultaneous training of two neural networks, with one serving as a discriminator and the other acting as a generator. The primary role of generator is to produce fake information that accurately reflects the attributes of genuine training dataset. In contrast, the discriminator is responsible for discerning the differences between original and fake information. The collaborative interactions inherent within these networks contribute to an enhancement of their performance, facilitated by competitive interactions occurring all through the training procedure. The generator aims to delineate the distribution of samples obtained from an unknown data source, referred to as p_{data} , as demonstrated by the training dataset. Beginning with a foundational latent distribution, represented as p_z , the generator performs a transformation to produce $p_g = G(p_z, \gamma)$. The aim is for p_g to achieve convergence with p_{data} , a condition that is rarely satisfied beyond the simplest cases. This study introduces a conditional vector γ that encompasses the absorption spectra in conjunction with random noise data. The discriminator (D) and generator (G) engage in a minimax game, in which the generator aims to minimize the likelihood that the discriminator can correctly differentiate between real and synthetic data, while the discriminator seeks to maximize this likelihood. The value function pertinent to this game is delineated as follows [15]:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x, \gamma)] + \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z, \gamma)))] \quad (1)$$

- $\mathbb{E}$: Expected value
- γ : Conditional vector (Absorption Spectra)
- $x \sim p_{\text{data}}(x)$: Real sample from the training distribution
- $z \sim p_z(z)$: Latent noise sample from the prior
- $G(z, \gamma)$: Generated data conditioned on γ
- $D(G(x, \gamma))$: Discriminator output, estimated probability that input u is real (conditioned on γ)

The neural networks engage in a process of iterative training, wherein each network is dedicated to the minimization of a specific loss function.

$$\begin{aligned} \mathcal{L}_D &= -[y \log D(x, \gamma) + (1 - y) \log (1 - D(G(z, \gamma)))] , \\ \mathcal{L}_G &= -\log D(G(z, \gamma)). \end{aligned} \quad (2)$$

Within this framework, the variable y serves as an indicator for binary classification, where a value of $y = 1$ signifies authentic data, and a value of $y = 0$ indicates synthetic data. $\mathcal{L}_D$ denotes the discriminator loss and $\mathcal{L}_G$ the generator loss.

The proposed formulation of $\mathcal{L}_G$ represents a departure from the original setup of Generative Adversarial Networks (GANs), which aims to minimize $\log[1 - D(G(z, \gamma))]$. This specific methodology has been recognized for producing insufficient gradients [14, 48].

We would like to note that deep-learning based methods (e.g., cDCGAN) have demonstrated the feasibility of joint material–geometry inverse design for metasurfaces, though such methods remain limited by high-dimensional pixel-space representations, large data requirements, and difficulties in exploring beyond the training manifold [15]. Our work proposes a hybrid quantum-classical framework known as Latent Style-based Quantum GAN (LaSt-QGAN) that integrates a pretrained latent-space autoencoder (VAE/IWAE) with a style-based quantum generator, enabling efficient sampling from a compressed design manifold [39, 43, 49]. This latent-style QGAN yields substantial reductions in data and computational cost, maintains material-geometry co-design flexibility, and – importantly – demonstrates generation of high-Q, narrow-band metasurface designs that lie outside the original training distribution. To the best of our knowledge, this is the first quantum-augmented generative inverse-design pipeline applied to metasurfaces (or nanophotonic devices) that unifies latent compression, quantum sampling, material substitution, and practical manufacturability in a single end-to-end workflow.

2 Methods

The current work investigates the inverse design methodology, focusing on two specific types of absorbing metasurfaces, as depicted in Fig. 1 - hybrid dielectric metasurfaces and metal-insulator-metal (MIM) configurations. The configuration of metal-insulator-metal (MIM) structures is characterized by the presence of a dielectric layer between two metallic components. The examined structures exhibit a detailed Lorentzian-shaped absorption response, making them particularly effective for tasks concerning thermal radiation and energy harvesting [50, 51]. Hybrid dielectric metasurfaces utilizing a dielectric resonator over a metallic film substrate exhibit cavity effects that facilitate the generation of an asymmetric, narrow-band Fano resonance. This characteristic renders them especially advantageous

Fig. 1 Metal-insulator-metal (MIM) and hybrid dielectric metasurface unit cells demonstrate distinct absorption spectra characterized by Lorentzian and Fano shapes, correspondingly

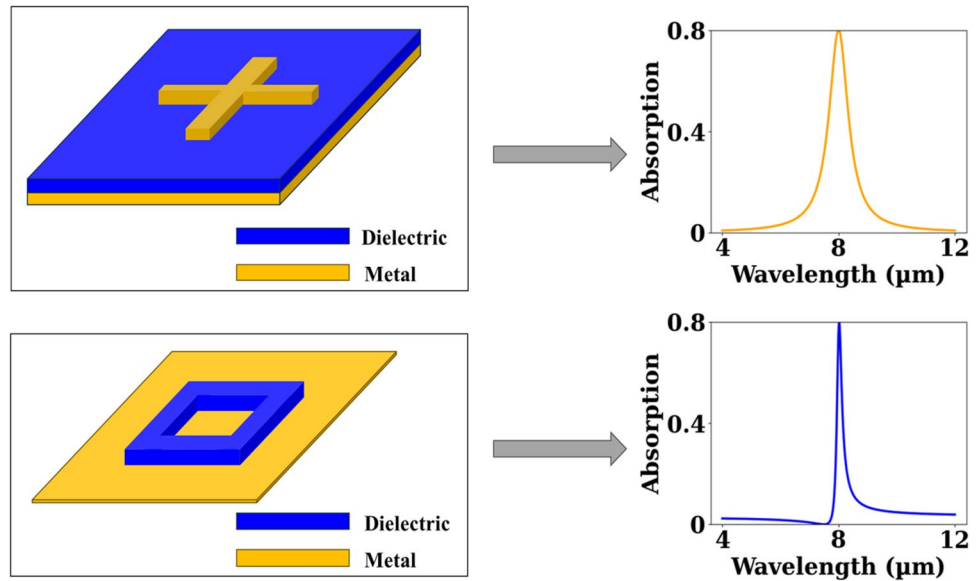
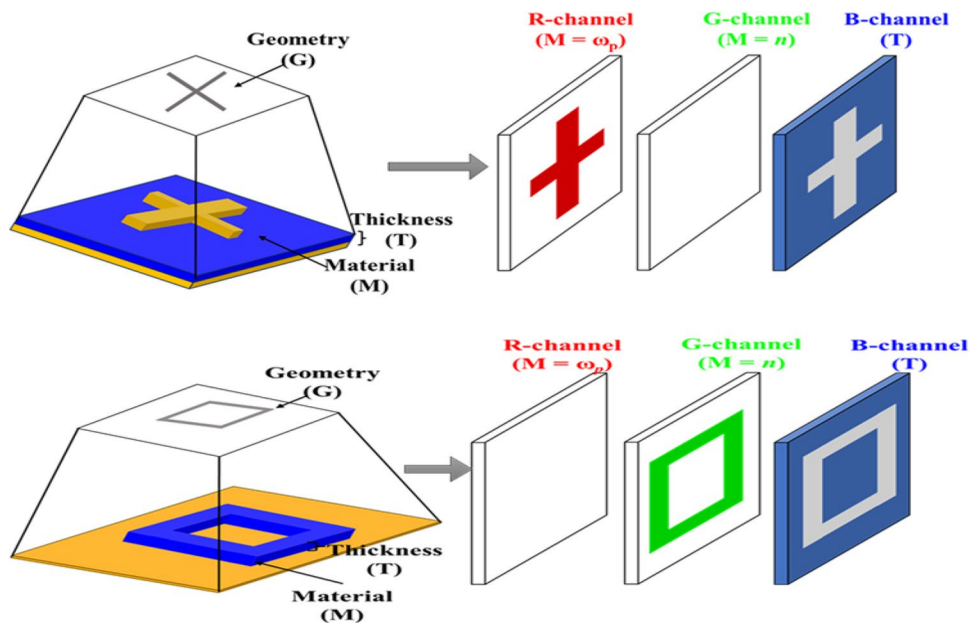


Fig. 2 RGB encoding of metasurface unit cells, where geometry is embedded and material properties, thickness, and structural features are mapped to individual color channels



for sophisticated functions in the field of photonics such as sensors [52].

We would like to note here that in conventional plasmonic resonant metasurfaces, ohmic losses typically limit Q to below ~ 10 , whereas early all-dielectric Fano/EIT-like metasurfaces experimentally achieved $Q \approx 483$ [53]. Quasi-BIC dielectric metasurfaces have demonstrated experimentally measured Q factors on the order of 10^3 – 10^4 , with reports reaching $Q \approx 5,000$ [54]. Very recent low-contrast BIC/quasi-BIC metasurfaces have achieved ultrahigh free-space accessible Q factors exceeding 10^5 , although these typically require careful control of radiative loss and fabrication tolerances [55, 56].

The encoding approach illustrated in Fig. 2 begins with the collection of planar topologies (G), material properties (M), and dielectric layer thicknesses (T) relevant to both MIM as well as hybrid dielectric metasurfaces. The parameters being examined are incorporated into the three separate channels of a colored image. The red channel is specifically associated with the plasma characteristics and the geometric configuration present within metal-insulator-metal (MIM) structures. The green channel demonstrates a correlation with the refractive index as well as the geometrical arrangement of dielectric elements within hybrid structures. The evaluation of the dielectric layer's thickness is conducted via the blue channel for both categories. For MIM structures, the encoding produces a red-blue color

spectrum, whereas for hybrid dielectric structures, it generates a green-blue color spectrum as shown in Fig. 2.

The dataset utilized for the absorption spectra training consists of 500 distinct metasurface unit cell layouts, represented as image-vector combinations. The structures presented in this study were derived from seven unique shape types, as outlined in the research conducted by Yeung et al 2021 [15, 57]. The investigation into the modeling of MIM and hybrid dielectric structures is performed utilizing unit cells with dimensions of 3.2×3.2 and $7.5 \times 7.5 \mu\text{m}^2$, respectively. The restricted availability of qubits poses a substantial constraint on our capacity to encode the complete set of values related to absorption spectra within the framework of quantum circuits. The parameters designated to represent the conditional vector are derived subsequent to the fitting of the Fano equation, specifically Resonance frequency (ν_0), Linewidth (Γ), peak absorption (A_0), and Fano asymmetry parameter (q) [58]. The parameters are articulated through a conditional vector denoted by (γ), which is further augmented by the inclusion of random noise (z). The design was transformed into a three-dimensional pixel representation, defined as a $64 \times 64 \times 3$ pixel "RGB" image, as illustrated in Fig. 2. We performed finite-element simulations of the generated designs using COMSOL Multiphysics to acquire an absorption spectrum vector range spanning from 4 to 12 μm [59].

Although the dataset used for LaSt-QGAN training consists of only 500 metasurface samples, the risk of overfitting is mitigated by the use of a pretrained variational autoencoder (β -VAE / IWAE). The autoencoder is trained on a substantially larger dataset (1500 metasurface images) to learn a smooth, low-dimensional latent manifold of physically valid metasurface geometries. During the subsequent QGAN training stage, this latent space is frozen, and only the quantum generator and classical discriminator are optimized. This significantly reduces the effective model capacity and prevents memorization of training samples. Moreover, the quantum generator operates as a latent-space sampler rather than a pixel-space generator, enabling stochastic exploration of the latent manifold and effectively acting as a quantum-assisted data augmentation mechanism. The ability of the model to generate metasurface designs with Q-factors exceeding those present in the training dataset further indicates strong generalization beyond simple interpolation.

It is worth noting that while 500 labeled samples would be critically insufficient for learning a direct mapping in the high-dimensional pixel space ($64 \times 64 \times 3$ dimensions), our hybrid approach fundamentally alters the learning landscape. By freezing the pre-trained VAE, which was trained on a larger dataset of 1500 geometries, the QGAN is tasked only with learning the probability distribution within

a compressed 9-dimensional latent space. In this reduced dimensionality, 500 spectral-geometric pairs provide a sufficient density of data points to stabilize the quantum generator, effectively mitigating the of dimensionality that plagues classical pixel-space GANs.

In the color-encoding phase the plasma resonance frequencies in metallic metasurfaces are modeled by the Drude model and encoded as the red channel. The plasma frequencies observed in this study are quantified as follows: for gold (Au), the plasma frequency is measured at $\omega_P = 1.91$ PHz; for silver (Ag), it is $\omega_P = 2.32$ PHz; and for aluminum (Al), the plasma frequency is recorded at $\omega_P = 3.57$ PHz. The refractive indices of the dielectric resonators are measured for the green channel, yielding values of $n = 2.4$ for zinc selenide (ZnSe), $n = 3.46$ for silicon (Si), and $n = 4$ for germanium (Ge). The optical constants pertinent to the mid-infrared wavelength spectrum employed in the modeling were utilized to derive the material features. The blue channel represented the dielectric thickness, in the range 100 – 950 nm. In order to maintain the integrity of the "RGB" color system, a normalization process was applied to all encoded values, ensuring they fell within the specified range of 0 to 255.

3 Architecture

The comprehensive workflow for both training and inference processes associated with LaSt-QGAN is visually represented in Figs. 3 and 4, respectively. Figure 3 illustrates training architecture of LaSt-QGAN. During training, images provided undergo transformation via Variational Autoencoders (VAE) to effectively encode essential data in a dense format, thereby accommodating the constraints of resource-limited Noisy Intermediate-Scale Quantum (NISQ) computers, rather than engaging in a pixel-by-pixel learning approach. Figure 4 illustrates the application of the decoder from the pretrained Variational Autoencoder (VAE) in generating images of the metasurface structures. This happens when the fully-trained quantum generators are used after the LaSt-QGAN has been trained. The Python script was executed on a GPU server, NVIDIA DGX-1, which is equipped with 1 TB RAM, 15 TB SSD storage, 128 CPU cores, and a single A100 (Volta) GPU, operating on Ubuntu 20.04 operating system.

3.1 Variational autoencoder

During the first step, a standard convolutional variational autoencoder (VAE) is used to get compressed features (f) from Input Metasurface images (I) that are in a latent field with a certain number of dimensions. The Variational

Fig. 3 Schematic Training Architecture of a Hybrid Quantum Generative Adversarial Network (QGAN) with a Pretrained Variational Autoencoder (VAE)

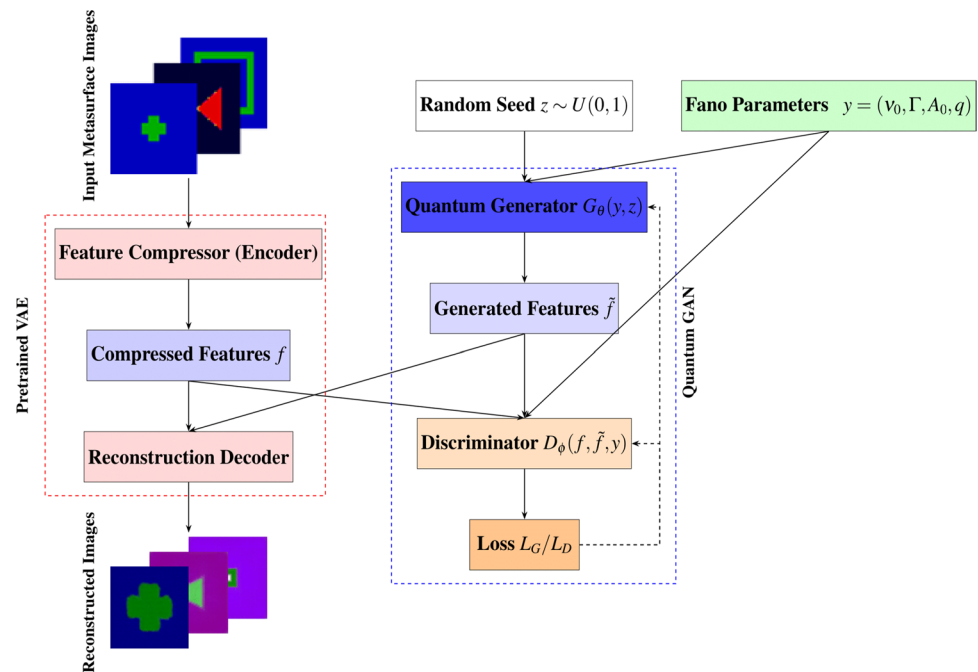
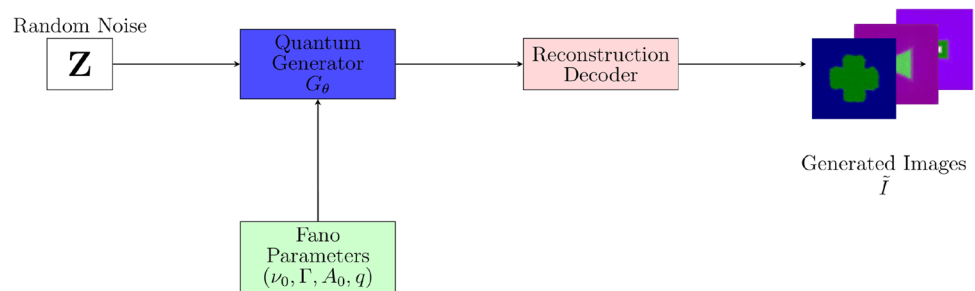


Fig. 4 Testing and Validation Process of a Hybrid Quantum GAN: From Latent Space Sampling to the Generation of Synthetic Metasurface Structure Images



Autoencoder (VAE) goes through the pre-training phase using the original image dataset, which has $64 \times 64 \times 3$ pixels. This process can be used to reduce the number of dimensions. The derived features make up the basic training dataset that the current quantum GAN learning process uses.

We perform a comparative analysis of the outcomes derived from two pretrained variational autoencoders: the β -VAE and the Importance Weighted Autoencoders (IWAE) [60, 61]. Unsupervised neural network architectures β -VAE and IWAE are created within the PyTorch framework with the goal of streamlining data compression and dimensionality reduction procedures [62]. The primary distinction between β -VAE and IWAE lies in how they learn the latent features of data. The β -Variational Autoencoder (VAE) introduces a control parameter, denoted as β , which serves to modulate the trade-off between the fidelity of data reconstruction and the organization of hidden features within the latent space. This adjustment enhances the interpretability of the extracted features. The Importance Weighted Autoencoder (IWAE) employs a technique known as importance

sampling, which enhances its ability to accurately represent intricate data patterns. In the conducted simulations, the parameter β is considered to be 2 in both the models, while for the Importance Weighted Autoencoder (IWAE), the number of samples is considered to be 5. Figure 5 demonstrates that both Variational Autoencoders (VAEs) function by transforming high-dimensional data into a lower-dimensional latent space and the decoder reassembles data from the latent representations (f).

3.2 Quantum generator

The Quantum Generative Adversarial Network (QGAN) operates as a generative framework, utilizing a quantum circuit to produce synthetic attributes derived from random noise in conjunction with a conditional vector (γ). In parallel, a conventional discriminator is assigned the role of differentiating between genuine characteristics (f) obtained from the β -Variational Autoencoder (VAE) or Importance Weighted Autoencoder (IWAE) and fake features ($\tilde{f}$).

Fig. 5 Architecture of the β -VAE and IWAE used in the paper

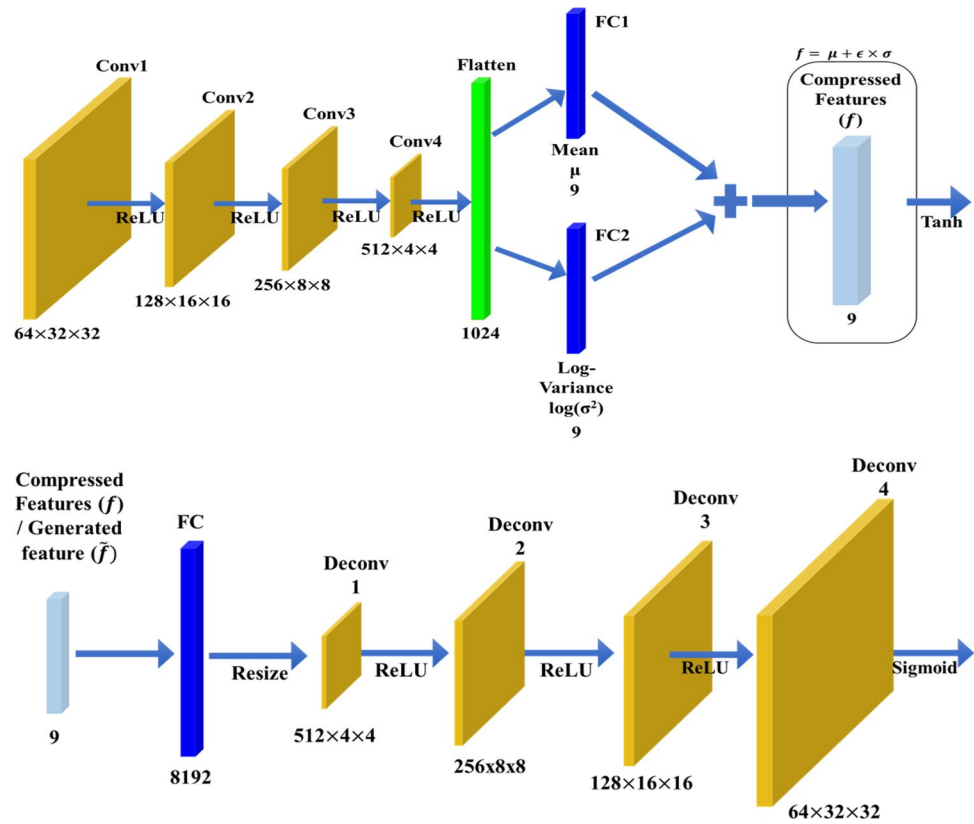
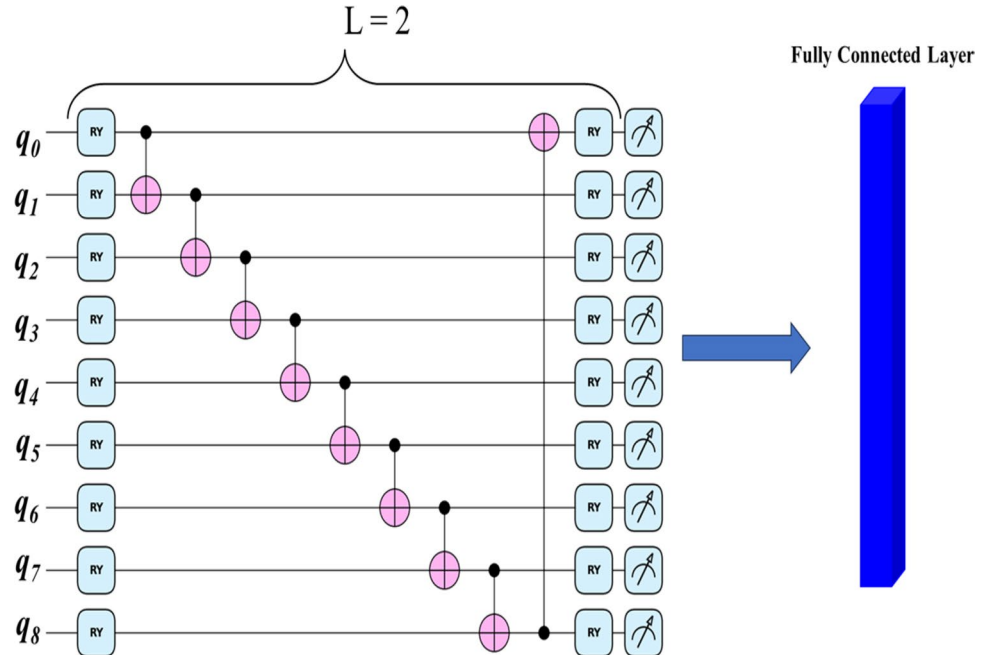


Fig. 6 Schematic of the quantum neural network employed for the LaSt-QGAN model. The circuit utilizes 9 qubits and $L = 2$ layers. It takes the input from the Fano parameters and the random noise as per (3) and gives the output to the Fully connected layer as shown in Fig. 3



generated by the quantum generator, implemented via Xanadu Technology's PennyLane simulator [63].

Figure 6 illustrates a quantum generator constructed using a style-based architecture [64]. In this framework, the rotational parameters within the training layers are influenced

by latent data. The system is designed to process input data through the concatenation of the latent noise vector (z) and conditional information (γ). This integration enhances the learning process, making it more dynamic and adaptable. The composite is subsequently channeled through a series

of sub-generators, which are implemented as variational quantum circuits [65, 66]. During training, the generator is progressively optimized using circuits with adjustable parameters, θ , which are altered throughout the training process. The quantum circuit consists of parametrized layers of rotation gates RY & entanglement layers CNOT gates. Here we use RY gates to limit probability amplitude to real-valued. CNOT gate entangle every q_i with its neighbor q_{i+1} (and q_N with q_0) for maximizing the effective entanglement while minimizing the gate count and susceptibility to decoherence. The current implementation features a quantum circuit structured with parameterized layers that include rotation gates identified as RY. Circular entanglement is then added to the configuration by using CNOT gates. Figure 6 uses a 9-qubit circuit, with $L = 2$ layers, each consisting of $RY(\theta)$ rotations followed by circular CNOT entanglement and then again $RY(\theta)$ gate. Input is encoded using angle encoding with affine-transformed parameters as shown in (3). The comprehensive unitary transformation U of the circuit is given by $U = U_i(z, \gamma) \cdot \dots \cdot U_1(z, \gamma)$. In this formulation, each layer i of the circuit integrates an orthogonal transformation that is applied to the input x .

$$\theta_i = W_i \cdot x + b_i, \quad (3)$$

where,

- W_i : The orthogonal weight matrix of size $n_{qubits} \times n_{qubits}$ with n_{qubits} being the number of qubits [67].
- b_i : The bias vector of size n_{qubits} .
- $x : [z|\gamma]$ is the concatenation of latent noise (z) and conditional vector (γ).
- θ_i : The rotation angles for the quantum gates in the i -th layer of the quantum circuit.

Each circuit uses the affine transformation described in (3) to incorporate the input (x) into the rotation angles (θ_i) corresponding to the quantum gates. In order to improve accuracy and efficiency and address problems with disappearing or inflating gradients in quantum neural network designs, the weight matrices are orthogonalized. Moreover, these matrices are evenly distributed in the interval $[-0.01, 0.01]$ [67]. The bias b_i is randomly selected from a uniform distribution in the range -0.01 to 0.01 .

During training, the model systematically optimizes the parameter set ($\Theta = \{W_i, b_i\}$) across all layers, iteratively adjusting the angles to enhance the generative capacity of the quantum model. The results obtained from the quantum circuit are expressed in terms of Pauli-Z expectation values, $\langle \sigma_z \rangle$. The aforementioned process is systematically replicated across all sub-generators, with each one playing

a significant role in the overall output. In order to maintain consistency, the latent space is constrained to the interval $[-1, 1]$, which corresponds to the output range of $\langle \sigma_z \rangle$ produced by the quantum generator. This framework enables the representation of complex, high-dimensional information, making it suitable for a variety of tasks in the field of quantum machine learning.

3.3 Classical discriminator

This study utilizes a classical discriminator, which is a multilayer perceptron (MLP) consisting of fully connected neural network, designed to distinguish between the latent space (f) and the synthetic feature representation ($\tilde{f}$) in the context of the LaSt-QGAN. The model utilizes input features in conjunction with an equivalent conditional vector (γ), which are subsequently processed through a series of layers as shown in Fig. 3. At the outset, the conditional vector data is subjected to a linear transformation in order to align its dimensionality with that of the input features. The classical discriminator is composed of three fully connected layers. The initial layer contains 128 nodes, which is succeeded by a second layer featuring 64 nodes. Subsequent to each of these layers, a Leaky ReLU activation function is applied, characterized by a negative slope of 0.2 [68]. The concluding layer consists of a single node, followed by a Sigmoid activation function [69]. This configuration produces a probability score that indicates the genuineness of the input data, thereby assessing whether it is real or synthetic. The feedback process plays a vital role in directing the quantum generator to enhance the realism of its output features.

4 Results and discussions

4.1 LaSt-QGAN validation

The values obtained for plasma frequency (ω_p) and refractive index (n) are utilized to obtain new materials from LaSt-QGAN based metasurface. It is not possible to manufacture the materials using material properties generated by inverse design of metasurface. Hence, material properties are incorporated into the model, enabling the model to predict diverse characteristics that might otherwise be missed due to oversimplified categorization. To evaluate the effectiveness of the trained La-St QGAN and the applied image processing techniques, a set of Fano parameters is introduced alongside arbitrary latent vectors.

Figure 7 illustrates a sequence of tests utilizing inputs derived from the validation dataset, constituting 10% of the training data. The red lines indicate the inputs that have

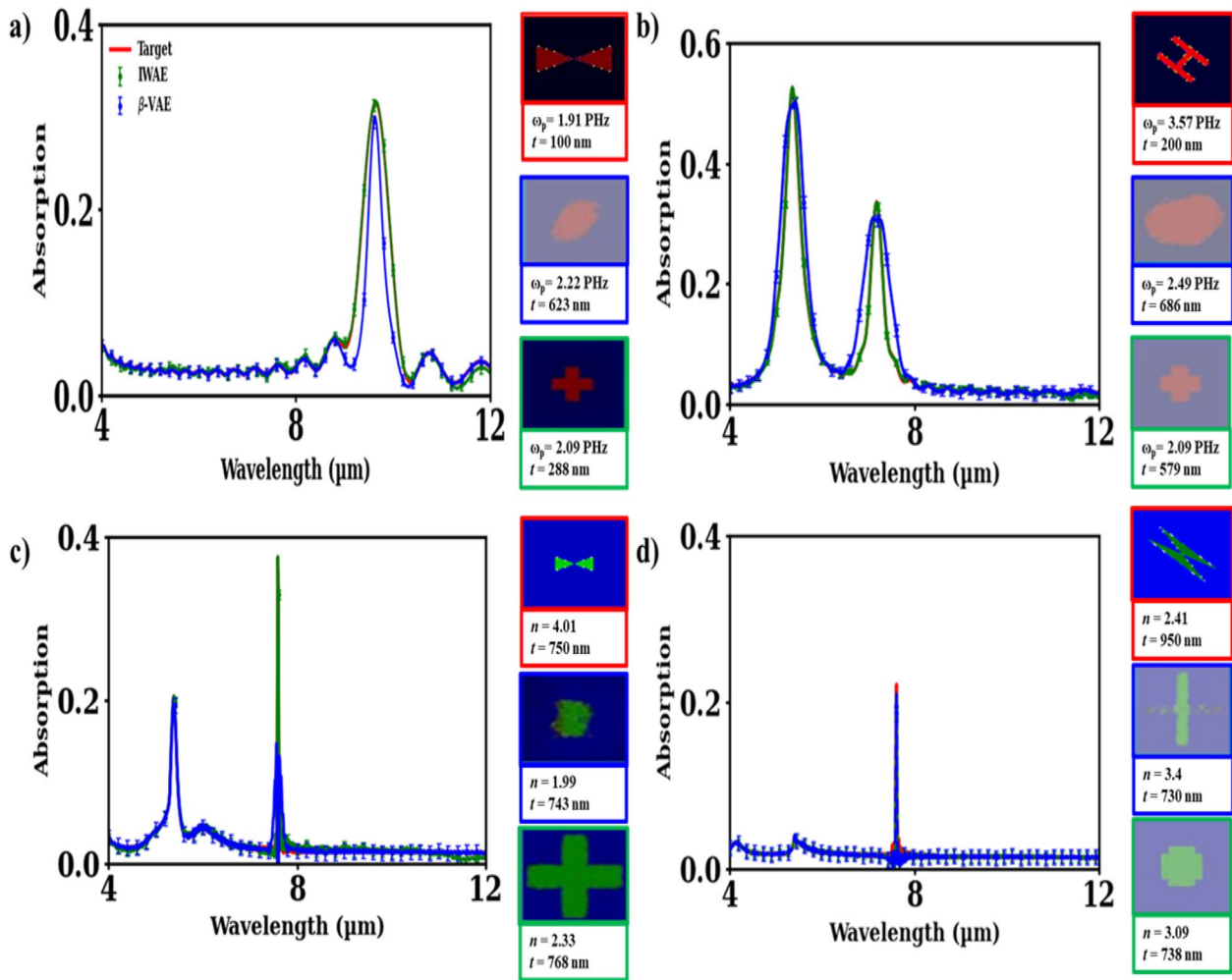


Fig. 7 Absorption metasurface for both **a, b**) Lorentzian-shaped and **c, d**) Fano-shaped spectra from the validation dataset (red), randomly selected as input targets for the La-St QGAN. Simulated spectra from

La-St QGAN designs (blue and green) using pretrained β -VAE and IWAE respectively are compared to the targets

been randomly selected for the study of absorption metasurfaces. The blue and green lines represent the simulated spectra associated with designs produced by La-St QGAN, employing pretrained β -VAE and IWAE methodologies, respectively. The corresponding unit cells are illustrated alongside each plot. To quantify the stability and reproducibility of the generative process, we performed a statistical analysis using $N = 100$ independent inverse design realizations for each target spectrum. Variance in resulting spectral responses obtained from the distribution of inverse designs is indicated as error bars in the Fig. 7. It is observed that variance is ± 0.006 for IWAE-generated spectra and ± 0.01 for β -VAE-generated spectra, highlighting the comparatively higher reconstruction uncertainty associated with the β -VAE framework. Based on the comparative validation presented in Figure 7, the IWAE-based architecture demonstrated superior reconstruction fidelity with significantly lower spectral deviation (± 0.006) compared to the β -VAE

(± 0.01). In the context of inverse design for high-Q metasurfaces, spectral accuracy is paramount; even minor reconstruction errors can lead to the loss of resonant features. Consequently, the IWAE framework is selected as the generative backbone for the subsequent material substitution (Section 4.2) and high-Q resonance experiments (Section 4.3).

4.2 Material Substitution

To improve the manufacturability and practical applicability of the proposed framework, we introduced the integration of LaSt-QGAN in conjunction with a publicly accessible database [70]. Figure 8 shows how the proposed substitution replaces estimated material properties with the most closely aligned characteristics derived from conventional materials. Figure 8a and 8b illustrates the application of this method on Lorentzian-shaped spectra, where the expected materials are

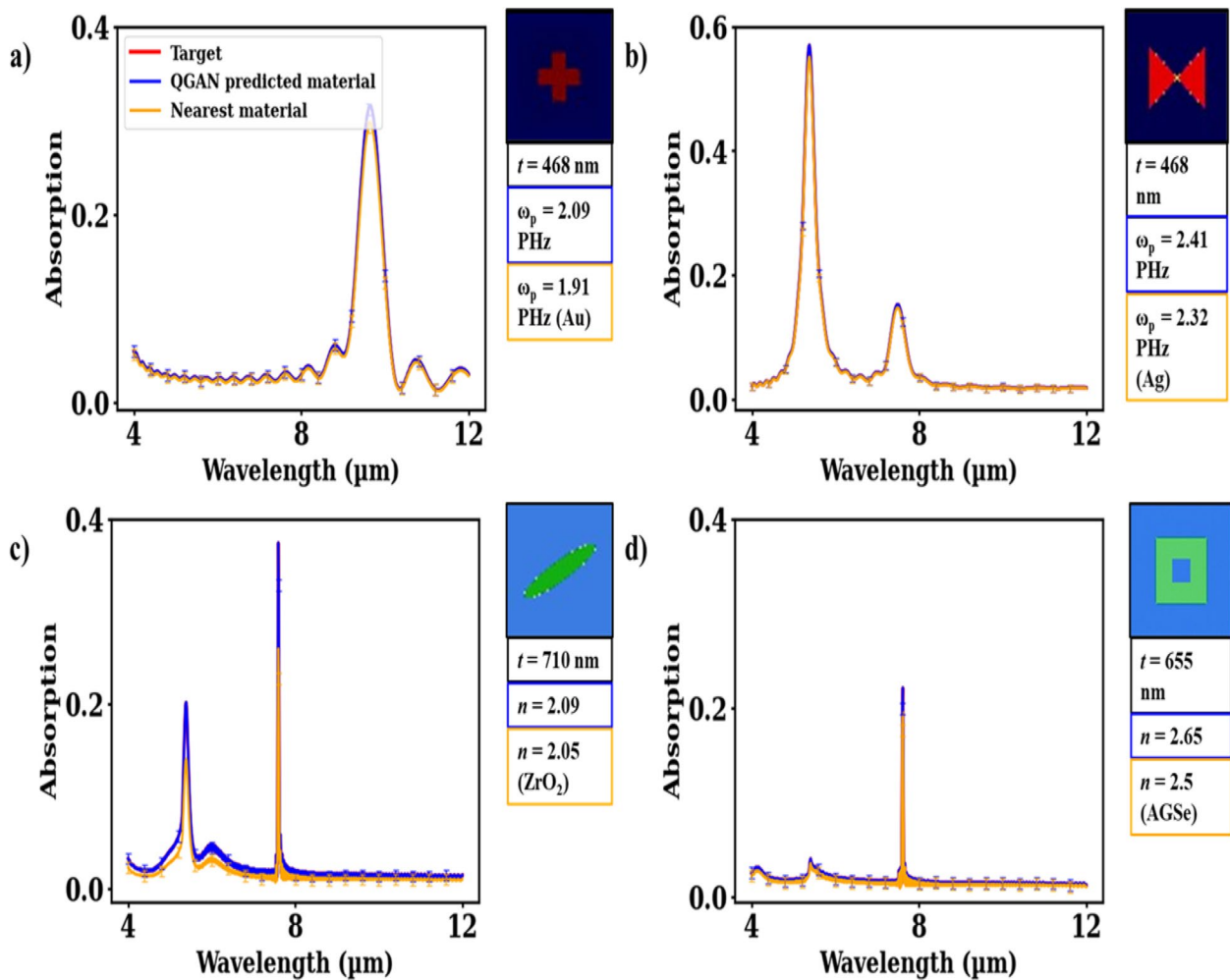


Fig. 8 Substitution of LaSt-QGAN-predicted materials with similar alternatives for absorption metasurfaces. **a, b**) Lorentzian-shaped and **c, d**) Fano-shaped target absorption spectra (red curves) are compared

replaced with traditional metals such as silver (Ag) and gold (Au) respectively. Forward simulations of these look-up table exhibit 95% precision as compared to the results produced by the LaSt-QGAN. Precision was quantified using normalized spectral overlap $S = \frac{\int A_{GAN}(\lambda) A_{sub}(\lambda) d\lambda}{\sqrt{\int A_{GAN}^2 d\lambda \int A_{sub}^2 d\lambda}}$, yielding $S = 0.95$ (95%). Here, A_{GAN} denotes ideal model-predicted absorption spectrum and A_{sub} denotes absorption spectrum after practical material substitution. Error bars in the Fig. 7 show variance of ± 0.006 calculated for about 100 inverse design realizations. The simulated structures show near perfect alignment with the target responses.

Figure 8c and 8d presents results of input targets characterized by Fano-shaped spectra. The LaSt-QGAN enables the prediction of a range of geometries, with corresponding refractive index values of $n = 2.09$ and 2.65 , respectively. The materials used for substitutions are zirconium dioxide (ZrO_2 , $n=2.05$) and silver gallium selenide ($AgGaSe_2$,

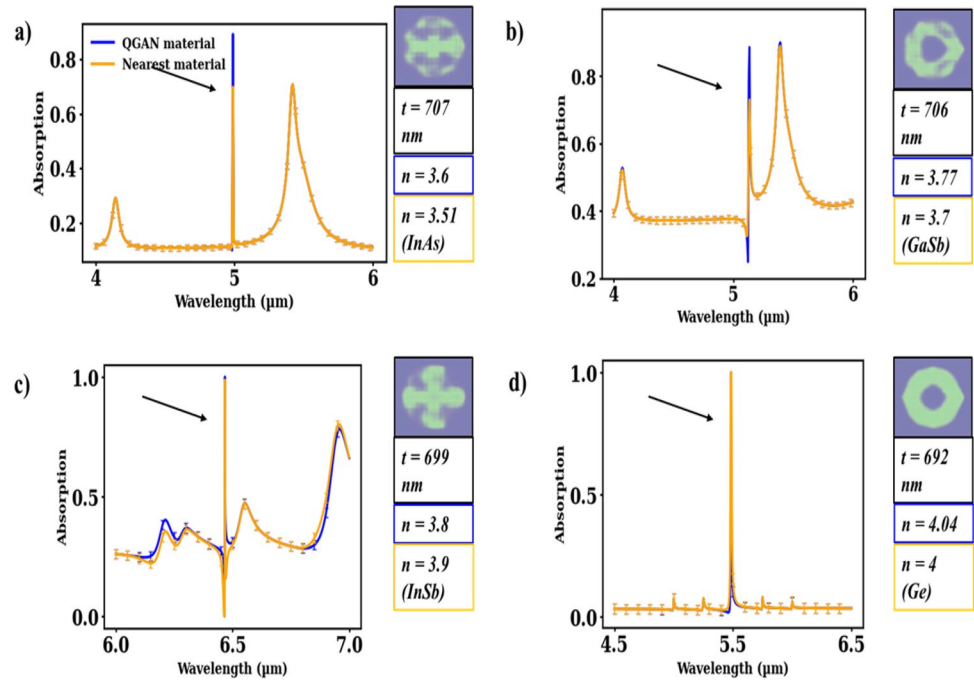
against simulated outcomes with nearest known materials (orange curves) and original GAN-predicted materials (blue curves)

$n=2.5$). Notably, certain materials identified through this process, including ZrO_2 and $AgGaSe_2$, were absent from the initial training dataset. This highlights the ability of the LaSt-QGAN to forecast new material parameters, including refractive index or plasma frequency, which are essential for identifying alternative materials that satisfy the target requirements. This hold true even in cases where these materials were not included in the training dataset.

4.3 High-Q resonances

The absorption spectra of the inversely designed metasurfaces exhibit pronounced high-Q Fano resonances as shown in Fig. 9. Figure 9(a – d) correspond to Q-factors of 1.38×10^4 , 2.0×10^4 , 1.2×10^4 , and 0.95×10^3 , respectively at the target wavelengths of $4.99 \mu m$, $5.13 \mu m$, $6.467 \mu m$, and $5.484 \mu m$. These high-Q resonances result from the engineered interference between discrete resonant

Fig. 9 Comparison of absorption spectra for inverse-designed metasurfaces exhibiting High-Q Fano resonances. The blue curves represent the ideal QGAN-predicted material response, while the orange curves show the performance after substituting the predicted refractive index with the nearest real-world material



modes and the continuum background, producing the characteristic asymmetric Fano line shapes.

To rigorously evaluate the model's generalization capabilities on out-of-distribution (OOD) targets, we tasked the LaSt-QGAN with generating metasurfaces exhibiting optical properties significantly exceeding those found in the training data. Specifically, while the training dataset was limited to structures with Q-factors of the order of 10^3 , we introduced target conditional vectors requesting Ultra-High-Q resonances ($Q \approx 10^4$).

Figure 9 also shows the extrapolated responses in the range of 0.95 to 2×10^4 . The ability to extrapolate order-of-magnitude improvements in the Q-factor indicates that the quantum generator has learned the underlying physics-governed manifold of the design space, rather than merely memorizing or interpolating the training examples. This constitutes a robust validation of the model's performance on truly out-of-distribution spectral targets.

The absorption spectra shown in Fig. 9 include error bars corresponding to a deviation of ± 0.01 , accounting for spectral variability across 100 inverse-design realizations in the high-Q regime. It is observed such high Q-resonances are a result of asymmetry in the geometries. In asymmetric geometries degeneracy in resonant modes are broken and radiative losses are suppressed, thereby sharpening the responses. The inset images in Fig. 9 depict the corresponding freeform metasurface unit cell designs that produce these high-Q Fano resonances. The emergence of asymmetric geometries from the model highlights its ability to transcend the initial design constraints and explore a broader, more complex parameter space, enabling the discovery of

non-intuitive structures that maximize resonance sharpness and spectral selectivity.

While the normalized spectral overlap (S) provides a broad measure of similarity, it may not fully capture the degradation of resonance sharpness in high-Q regimes, where small spectral shifts can impact performance. To assess this, we explicitly compared the quality factors before and after material substitution (Q_{ideal} vs. Q_{sub}). For the high-Q designs ($Q \sim 10^4$), substituting ideal material parameters with the nearest database matches resulted in Q-factor deviations of 1.91%, 3.37%, 6.67%, and 4.55% for the structures in Fig. 9(a)–(d), respectively. This analysis indicates that while the discrete nature of available refractive indices introduces minor spectral detuning, the topology generated by the LaSt-QGAN is sufficiently robust to absorb these variations. Crucially, even in the worst-case scenario, the substituted designs successfully maintain Q-factors within the 10^4 order of magnitude, ensuring that practical manufacturability is achieved without compromising the device's functionality.

4.4 Quantum hardware

To assess the performance of the quantum generator on actual hardware, the QGAN model, initially trained on a simulator, is validated on IBM Quantum `ibm_fez` superconducting Heron quantum processor through the PennyLane–Qiskit interface. The hardware tests were conducted with shots set to 4096, an optimization level of 3, and a resilience level of 2. This configuration allowed for the application of advanced transpilation techniques and medium-level

error mitigation strategies, such as zero-noise extrapolation and gate twirling, to minimize errors caused by decoherence [71, 72].

- `optimization_level = 3`: This option activates the most assertive transpilation mode, where the compiler puts in maximum effort to minimize circuit depth, gate count, and routing overhead, including advanced gate-block resynthesis and the use of multiple layout/routing seeds.
- `resilience_level = 2`: This facilitates a moderate degree of error mitigation, enhancing readout error mitigation and measurement twirling through Twirled Readout Error eXtinction (TREX) and incorporating supplementary approaches such as zero-noise extrapolation (ZNE) and gate twirling.

To address circuit-induced errors, we utilized the highest level of compilation effort available, along with a highest error-mitigation protocol possible, that considered both measurement and gate noise. This setup ensured an optimal balance between computational overhead and accuracy, allowing for reliable execution on actual quantum hardware. The Pauli-Z expectation values obtained from the device were then processed through the pre-trained IWAE decoder to generate candidate metasurface designs, which were evaluated using the same physics-informed performance criteria applied during simulation.

To evaluate the practical applicability of the proposed system on physical hardware, we analyzed the material substitution capabilities using the IBM Quantum `ibm_fez` processor. To quantify the stability of the generative process on NISQ hardware, error bars shown in Fig. 10 show a variance of ± 0.006 calculated for $N = 30$ independent hardware executions for each target spectrum. Figure 10(a)

and (b) illustrate the application of this method on Lorentzian-shaped spectra, where the hardware-predicted material properties are replaced with the most closely aligned characteristics derived from conventional materials. For the metallic resonator shown in Fig. 10(a), the LaSt-QGAN predicted a plasma frequency of $\omega_p = 3.26$ PHz, which is successfully substituted with Beryllium (*Be*, $\omega_p = 3.33$ PHz). Similarly, Fig. 10(b) demonstrates that the hardware-generated design yield a refractive index of $n = 3.26$, exhibiting a high degree of alignment with Gallium Arsenide (*GaAs*, $n = 3.3$). These results confirm that the quantum generator maintains the precision required to identify alternative materials that satisfy target requirements, even under the influence of hardware noise.

To rigorously evaluate the model's generalization capabilities on out-of-distribution (OOD) targets using NISQ hardware, we tasked the quantum processor with generating metasurfaces that exhibit Ultra-High-Q resonances ($Q \approx 10^4$). Figure 10(c) and (d) present the results of this extrapolation test observed from the $N = 30$ hardware realizations, confirming that the model successfully generates geometries supporting sharp asymmetric line shapes despite the limited training data ($Q \sim 10^3$). The hardware-predicted refractive indices of $n = 3.6$ and $n = 4.04$ are substituted with Indium Arsenide (*InAs*, $n = 3.51$) and Germanium (*Ge*, $n = 4$), respectively. This shows the ability to extrapolate order-of-magnitude improvements in the Q-factor on a physical quantum device.

4.5 Comparative analysis

Table 1 presents a comparative analysis between a conventional Generative Adversarial Network (GAN), as implemented in Yeung et al. [15], and the proposed Quantum GAN (LaSt-QGAN) framework for the inverse design of

Fig. 10 Experimental validation of LaSt-QGAN on IBM Quantum hardware for material substitution and high-Q resonance generation. a, b) Hardware-predicted material substitution matching Beryllium and GaAs for Lorentzian spectra. c, d) Generation of high-Q Fano resonances targeting InAs and Ge. The red curves represent target spectra, compared against hardware-predicted (blue) and nearest material (orange) outcomes

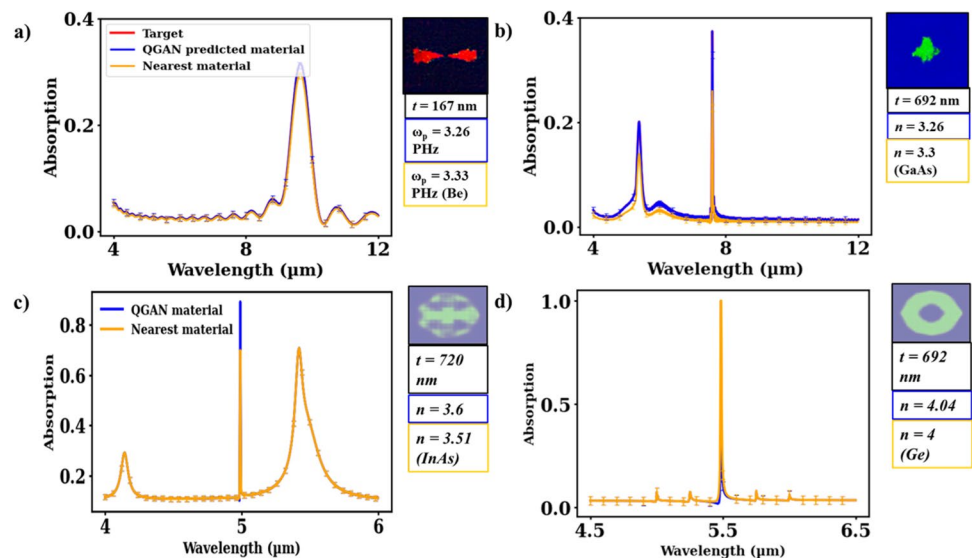


Table 1 Comparison of conventional pixel-space GAN and the proposed latent-space LaSt-QGAN performance in inverse design of metasurfaces.

Metric	DCGAN [15]	LaSt-QGAN (Pretrained IWAE)	LaSt-QGAN (Pretrained β -VAE)
Runtime (hrs)	25	4	4
MSE	10^{-3}	10^{-4}	10^{-3}
Training Samples	20,000	500	500

metasurfaces. The LaSt-QGAN significantly outperforms the classical GAN across all key performance metrics. Since VAE is pretrained we must account for the overhead of training time. The unsupervised pre-training of the VAE on 1500 geometry samples requires approximately 1.5 hours on the NVIDIA DGX-1 platform (A100 GPU). When combined with the LaSt-QGAN training time of 2.5 hours, the total runtime is approximately 4.0 hours. Comparing this to the baseline, the classical DCGAN required 25 hours of training to achieve convergence. Thus, even inclusive of the pre-training overhead, the proposed hybrid framework delivers a roughly 6-fold reduction in total computational time. Furthermore, it is critical to note that VAE pre-training is a one-time computational investment. Once the latent manifold of valid physical geometries is learned, the fixed VAE can be reused to rapidly retrain the quantum generator for diverse spectral targets or material constraints. In such recurring design scenarios, the marginal computational cost effectively drops to the 2.5-hour QGAN training time, exhibiting the 6-fold efficiency gain. Furthermore, the LaSt-QGAN achieves a mean squared error (MSE) of 10^{-4} , an order of magnitude lower than the GAN's MSE of 10^{-3} , indicating higher prediction accuracy with the target spectra. Notably, this improved performance is achieved using only 500 training samples, in contrast to the 20,000 samples required by the conventional DCGAN. For a fair comparison, both the classical GAN baseline and the proposed LaSt-QGAN are trained under identical optimization settings, including the same optimizer (Adam), learning rate of 0.001, batch size of 16, and number of epochs to be 500. The conditional inputs for both models are defined over the same spectral range (4–12 μm) with identical spectral resolution and parameterization.

These results underscore the effectiveness of the hybrid quantum inverse design framework in not only targeting specific resonance wavelengths but also enhancing resonance quality through subtle geometric modifications which is crucial for applications demanding narrow band absorption, enhanced sensing sensitivity, and precise control of electromagnetic wavefronts. It is important to note that the base model, a pixel-space Deep Convolutional GAN (DCGAN), works directly on the high-dimensional image data without compressing it. LaSt-QGAN, on the other hand, uses

a hybrid architecture where a pre-trained VAE shrinks the design space before the quantum generator is trained.

To accurately interpret the performance gains reported in Table 1, it is essential to decouple the contributions of the classical dimensionality reduction from the quantum generative modeling. The observed 6-fold reduction in training time is primarily attributable to the classical Variational Autoencoder (VAE) architecture. By compressing the high-dimensional design space ($64 \times 64 \times 3$ pixels) into a compact 9-dimensional latent vector, the VAE significantly reduces the computational complexity of the optimization landscape, a benefit that would apply to both classical and quantum latent-space models.

The 40-fold decrease in data requirements (training on 500 samples compared to 20,000) is a significant benefit that can be ascribed to Quantum Generator. Classical GANs notoriously require large dataset and are prone to mode collapse. We observed that quantum generator exhibits exceptional sample efficiency, adeptly learning the latent manifold topology from merely 500 samples, owing to the great expressibility of the Hilbert space. This capability enables the LaSt-QGAN to generalize and produce diverse, high-fidelity designs in a data regime where traditional deep classical GANs often fail to converge or simply memorize the training set.

4.6 Limitations

While the LaSt-QGAN framework demonstrates significant efficiency gains, several limitations need to be discussed to contextualize the results. Firstly, the smaller number of training samples may restrict the generation of geometry classes very distinct from original dataset, which in turn reduces the generalization of framework.

Second, the quantum generative training process was executed on PennyLane state-vector simulator which assumes an ideal, noise-free environment, disregarding the gate errors, decoherence, and shot noise inherent in physical NISQ hardware. Practical deployment would require the integration of quantum error mitigation (QEM) strategies to prevent gradient vanish (barren plateaus) and generative fidelity loss.

Third, the current framework operates under fixed constraints: specifically, fixed unit-cell dimensions ($3.2 \times 3.2 \mu\text{m}^2$ and $7.5 \times 7.5 \mu\text{m}^2$) and a defined spectral range (4–12 μm). The model in its current form is not scale-invariant and would require retraining to address different frequency bands (e.g., visible or terahertz) or pixel resolutions.

5 Conclusion

The proposed hybrid quantum-classical based on LaSt-QGAN method framework demonstrates significant improvements in inverse design of narrow band absorbing metasurfaces. The model optimizes metasurface structures for narrow-band absorption while also reducing computational costs through the integration of a Quantum Generative Adversarial Network (QGAN) with a Variational Autoencoder (VAE). It is observed that LaSt-QGAN method reduces training time by 10 times mostly owing to the VAE induced compression of sample space. It is also observed that data requirements are reduced by 40 times compared to traditional GAN-based methods. The model also achieves a low mean squared error (MSE) value of 10^{-4} with 500 samples in training dataset as compared to DCGAN which requires 20000 images to achieve 10^{-3} MSE value [15]. The method is implemented both on quantum simulator and QPU with error mitigation schemes.

Crucially, the LaSt-QGAN demonstrates robust out-of-distribution generalization capabilities. While the training dataset was strictly limited to structures exhibiting Q-factors on the order of 10^3 , the generative model successfully extrapolated beyond this learned regime to discover novel unit cell geometries supporting high-Q resonances ($Q \approx 10^4$). This ability to extrapolate order-of-magnitude improvements distinguishes the proposed generative framework from conventional interpolative methods, confirming that the quantum generator can effectively explore and identify high-performance designs that lie outside the boundaries of the original training distribution. These results are also replicated in QPUs using error mitigation strategies with similar generalization capabilities.

While the shallow depth of our ansatz ($L = 2$) is designed to be hardware-efficient, practical implementation would likely require the integration of Quantum Error Mitigation (QEM) techniques, such as Zero-Noise Extrapolation or Readout Error Mitigation, to preserve generative performance. Consequently, the efficiency gains reported here represent an algorithmic upper bound, and future work involving hardware-in-the-loop training is necessary to characterize the model's resilience to physical noise. Moreover, while this study prioritized the IWAE architecture for its superior spectral reconstruction, the β -VAE framework offers the unique advantage of learning disentangled latent representations. Future research could further explore this trade-off between fidelity and interpretability, analyzing the β -VAE latent space to correlate specific dimensions with physical properties (e.g., resonance linewidth or position), thereby enabling more intuitive, physics-driven navigation of the design manifold.

Future research can leverage the proposed latent-quantum generative pipeline to design adaptive metasurfaces whose responses remain stable under realistic environmental variability, including shifts in spectral content and incidence-angle distributions. Moreover, extending LaSt-QGAN to multi-objective optimization—for example, simultaneous control of absorption, phase and polarization, bandwidth, and angular tolerance—would support the inverse design of multifunctional devices and accelerate translation to applications in imaging, sensing, communications, and energy harvesting.

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Data Availability The data supporting the results in this work will be made available by the authors upon reasonable request.

Declarations

Conflicts of Interest Both the authors declare that they have no conflict of interest.

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